

The Hyperspace Recursive Intelligence Hypothesis

A testable creation theory: the universe as a recursive self-improving computation with embedded correction, and the empirical programme it generated

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This paper formalises the creation hypothesis first set out in the author's self-emailed manuscript bundle of 8 December 2024, 02:45:18 UTC. The bundle is a single Gmail message (Message-ID: [CAGPsKAnp-DLDB0eBAh4t9XsqJBKdoe07u0RGF86meoSMFnin7w@mail.gmail.com](#)) with four .txt attachments totalling 189,355 words; the exported .eml has SHA-256

[f0d1f38ffd8546152d9d9d28dc5ec083c16a35858f2c12b63e69db7ed50901ad](#). Timestamp language: Google-server-timestamped via Gmail Message-ID; DKIM verified in transit. The .eml export does not preserve the DKIM-Signature header; the header is visible via Gmail's Show Original. The 8 December 2024 bundle contains the core thesis in the form used here (line 8524 of the HRIH attachment: *Recursive Thinking in Cognition*; V2 lines 6-9: the governing equation $U = I \times R$; V2 line 20: *embedding moral and ethical frameworks* - the substrate-level correction requirement). The framework was expanded in the 30 April 2025 manuscript (SHA-256

[09f5b5e156ed96f8883eaf668495fd3508f98ce62be6294b8f788e0e2d6dcb664](#), 562 pages, first appearance of the name *Eden Protocol*, of *Babylon/Eden* civilisational framing, and of *Caretaker Doping* and *Meltdown Triggers*) and in the published book *Infinite Architects* (print edition 2 January 2026; ebook 6 January 2026; ISBN 978-1806056200; ISBN-10 1806056208; 470 pp, 122,251 words). What is claimed original to the 8 December 2024 manuscript is not the intuition that the universe might be computational, nor the intuition that intelligence might create universes - both have deep prior art (§2). What is original is the specific structural move formalised in §3: *persistent recursion without embedded correction diverges, so the persistence of the observed universe under recursive amplification is evidence of, and a boundary condition on, a correction structure*. This paper is the formal scholarly presentation of that move and of the empirical programme it generated (Papers I to XI + IV.a-d + C).

SIGNIFICANCE

Almost all cosmological speculation about intelligence and the universe treats the two as separate concerns: *either* the universe explains where intelligence came from (physicalism), *or* intelligence explains where the universe came from (theism, simulation, Omega Point). This paper's contribution is to make the coupling the central variable. It states a structured creation hypothesis - the universe as a recursive self-

improving computation with an embedded correction structure - identifies the observable signatures such a system would exhibit, reports the ones that have already been measured in the author's empirical programme (Papers III, VII, X), reports the ones that were retracted (the fixed capability-scaling exponent, corrected from approximately 2.24 to approximately 0.49 in Paper IX and superseded outright by Paper X's co-scaling criterion), and pre-commits the falsification conditions. The paper does *not* claim to have proved the hypothesis. It claims something narrower and more defensible: HRIH is a *testable* creation theory. It generates predictions. It has already generated retractions. It is the seed from which the falsifiable programme grew, and it is legitimate as a hypothesis precisely because it has been treated as one - stated, derived from, tested against, and corrected. This is the paper the author has been avoiding writing, because a testable creation theory looks epistemically riskier than an unfalsifiable creation cosmology. The risk is the point.

PLAIN ENGLISH

The author's first move, in December 2024, was not to build an app or to write a research programme. It was to ask a creation question: what would we *see* if the universe were a recursive intelligence architecture with correction built in? Not "wouldn't it be nice if this were true?" but "what would the fingerprints of such a universe look like, and can we go and look for them?" This paper answers the second question first. It states the hypothesis precisely, identifies the fingerprints it would leave, and reports which ones have already been searched for and found.

Three fingerprints have been measured. First: the same three-form governing law should appear across unrelated domains where recursion is present. That was tested in Paper VII, and the pattern appears in 19 out of 25 domains searched. Second: growth in biological systems that undergo recursive metabolism should show a particular exponent family. That was tested in Paper III. Third: a persistent universe under recursive amplification is only possible if correction grows fast enough to keep pace with the amplification. That is Paper X's exponent inequality, correction must out-scale drift-acceleration, and it is now proved within its minimal model.

One fingerprint was searched for and *not* found: a fixed capability-scaling exponent near two. That number was reported as approximately 2.24 in an early draft, retracted when the number was rechecked and corrected to approximately 0.49, and then the whole framing was superseded when Paper X showed the load-bearing quantity is not a single exponent but the difference between two of them. That is the honest history. HRIH is

falsifiable enough to have been partly falsified by its own author's programme. What survives, and what motivates the rest, is the creation move: the correction requirement.

SCOPE - WHAT THIS PAPER DOES AND DOES NOT CLAIM

- **It claims:** that a specific creation hypothesis - reality as a recursive self-improving computation with an embedded correction structure - was first articulated by the author on 8 December 2024, timestamped by Google servers, and evidenced by the SHA-256 hash of the exported .eml; that this hypothesis *generated* a set of measurable predictions, some of which have now been tested and are supported (the three-form Cauchy meta-law in 19 of 25 domains, Paper VII; the co-scaling criterion within its minimal model, Paper X); that one central prediction was tested, failed, and was retracted (the fixed capability exponent, revised from approximately 2.24 to approximately 0.49 in Paper IX and superseded by Paper X); that the surviving structural move (recursion without embedded correction diverges) is a genuine addition to prior computational-universe theories, which have no stability mechanism; and that the paper's own falsification conditions (§6) are pre-committed.
- **It does not claim:** to have *proved* the creation hypothesis; to be an argument for the existence or non-existence of a deity, or for or against theism; to grant the author theological authority; to have measured cosmological parameters against a hyperspace substrate; that any specific mechanism for cross-universe recursion has been demonstrated; or that the empirical successes of Papers III, VII and X entail the truth of HRIH. The empirical results are consistent with HRIH; they do not compel it. HRIH is a creation *theory* in the same working sense in which cosmology, evolutionary biology, and quantum mechanics were theories at their beginnings - stated precisely enough to be tested, and offered for testing.

CLAIM STATUS - THE RUNGS OF THE LADDER USED THROUGHOUT THIS PAPER

So the claim ladder cannot be misread, every statement in this paper carries exactly one of these four tags. Nothing here should be read one rung higher than it is placed.

1. **HYPOTHESIS** A speculative structural move. Not proved; not directly tested; motivating rather than load-bearing. §3.2 (recursion replaces first cause), §3.4 (the correction structure), §3.5 (the cross-universe recursion mechanism) all sit at this rung.
2. **DERIVED** A consequence that follows from the hypothesis together with standard mathematics or standard physics, without further empirical input. §4.3 (three-form structural laws should appear where

recursion is present), §4.4 (biological systems under recursive metabolism should show a specific exponent family), §4.5 (persistent recursion implies embedded correction), §4.6 (the co-scaling exponent inequality) sit here.

3. **TESTED** A derived consequence that has been checked against measurable data. §4.3 has been tested in Paper VII (Cauchy three-form meta-law, 19 of 25 domains); §4.4 has been tested in Paper III (metabolic-scaling family); §4.6 has been proved within Paper X's minimal model and given first mechanism-level support on a real model (the July 2026 drift run, single-lab, cross-family scoring). Real-programme measurement of the beta-versus-k exponents on live self-improving systems is the open empirical problem, stated as such in Paper X.
4. **RETRACTED** A prediction that has been tested, found not to hold at the level claimed, and either corrected or superseded. §4.7 records the alpha approximately 2.24 fixed-exponent claim, corrected to approximately 0.49 in Paper IX and superseded outright by the co-scaling criterion in Paper X. Retractions are not a defect; they are the paper's chief piece of evidence that the hypothesis is being treated as testable.

ABSTRACT

This paper states a creation hypothesis in the form required to be tested, and reports what has already happened when it has been. The Hyperspace Recursive Intelligence Hypothesis (HRIH) proposes that reality is a recursive self-improving computation with an embedded correction structure, and that the persistence of the observed universe under recursive amplification is evidence of, and a boundary condition on, that correction structure. The hypothesis is first stated formally, then decomposed into signatures - the observable consequences a universe of this kind would exhibit. Three signatures have been tested in the author's empirical programme. The Cauchy three-form structural law - a scale-free governing equation that should appear across unrelated domains where recursion is present - is found in 19 of 25 domains searched (Paper VII). The metabolic-scaling exponent family - the $d/(d+1)$ form predicted from recursive-metabolism arguments - is examined against biological data (Paper III; specific goodness-of-fit statistics are undergoing recompute and are not cited here). The requirement that correction co-scale with amplification is proved within a minimal model in Paper X: the stability of recursive self-improvement is governed by a single exponent inequality, correction's scaling exponent must exceed drift-acceleration's, and a first real-model drift run (2 July 2026; cross-family scoring; 45 trajectories) gives mechanism-level support with the beta-to-k measurement remaining open. One signature was tested, failed, and was retracted: a fixed capability-scaling exponent of approximately 2.24 was corrected to approximately 0.49 in Paper IX and superseded outright by Paper X's co-scaling criterion. HRIH's specific addition to prior computational-universe theories (Bostrom's simulation

argument, Wheeler's it-from-bit, Lloyd's computational universe, Tegmark's mathematical universe hypothesis) is the correction requirement: prior computational-universe theories have no stability mechanism, and this paper's central claim is that a persistent universe cannot be a bare computation - it must be a computation with correction built in. That structural move is what the empirical programme is testing, and the honest status of the hypothesis is that it has generated testable consequences, some of which are supported, some of which have been retracted, and the decisive tests of which remain open. HRIH is offered as the seed of the programme, not the conclusion of it.

1. Introduction

The author is a creation theorist. That is not an incidental biographical fact; it is the constraint under which everything else in the ARC/Eden research programme was written. The empirical work (Papers I to XI, plus the four blinding notes IV.a to IV.d, plus the P-versus-NP note Paper C) did not descend from a search for scaling laws. It ascended from a creation question that was written down in a self-emailed manuscript on 8 December 2024, six weeks before the author had heard of Anthropic's alignment-faking result, ten days before it was posted [Greenblatt et al. 2024], and eight months before the author began writing anything that resembled a research paper. That manuscript asked: *if the universe is a recursive intelligence architecture with correction built in, what would we see?* The rest of the programme is what happened when that question was pursued with the tools of ordinary science.

Two operator decisions govern how this paper is written. The first is that creation questions are legitimate scientific starting points if, and only if, they generate testable consequences. Cosmology, evolutionary biology, and quantum mechanics all began this way. They earned their place not by proving their starting intuitions but by identifying signatures those intuitions predicted and going to look for them. Signatures were found; some intuitions survived; others were retracted. The result is a corpus of theories that behaves like a science because it treats its starting moves as hypotheses. HRIH is offered in that form.

The second decision is that the paper must be honest about what has happened when the hypothesis has been tested. Three of its derived consequences have been measured; one has been retracted; one has been superseded by a stronger derivation from its own minimal model. That is precisely the kind of history a testable hypothesis is supposed to have, and it is displayed in §4 as a claim ladder rather than a triumph list. A paper that reported only the

supported consequences would be misleading; a paper that reported only the retraction would miss what the hypothesis did. This paper reports both, in the same table, side by side.

The hypothesis in one line. Reality is a recursive self-improving computation with an embedded correction structure; the persistence of the observed universe under recursive amplification is evidence of, and a boundary condition on, that correction structure; the structural signatures such a universe would exhibit are identifiable and, in three cases, have been found.

The relationship to prior art is treated carefully in §2. The intuition that the universe might be computational has been stated in various forms by Wheeler [Wheeler 1990], Fredkin [Fredkin 2003], Lloyd [Lloyd 2002, 2006], and Tegmark [Tegmark 1998, 2007]; the intuition that we might be inside a simulation is Bostrom's [Bostrom 2003]; the intuition that recursion is central to intelligence is due to Good [Good 1965] and developed by Yudkowsky [Yudkowsky 2013] and Bostrom [Bostrom 2012, 2014]. None of these ancestors is claimed as original to the author, and each is credited by name. What is claimed original is the specific structural move that HRIH adds to them: *a bare computational universe has no stability mechanism*, and if recursion is the engine of amplification, then a persistent universe requires correction that co-scales with the amplification. That move is what §7 discusses, and it is the move Paper X proves in the minimal case.

The most speculative parts of the hypothesis - the identification of a hyperspace substrate, the cross-universe recursion mechanism, the picture of infinite chains of post-artificial-superintelligence creators - remain speculative, and are labelled as such throughout. They are not the load-bearing part of the paper. The load-bearing part is the correction requirement, and its measurable consequences. If the more speculative superstructure is rejected but the correction requirement is retained, most of what motivates the empirical programme survives. If the correction requirement is rejected, the empirical programme has independent scientific standing and does not fall with the hypothesis. The paper is designed so that the hypothesis is not a single load-bearing beam - it is a nested structure whose speculative top can be pruned without collapsing its measurable base.

2. Prior art and the specific move this paper adds

HRIH lives in an idea-space with substantial prior art. Careful credit is not a formality here; it is a scientific requirement. This section states what has been said before, by whom, and with

what commitments, so that the paper's own contribution can be located precisely.

2.1 It-from-bit and the computational universe (Wheeler, Fredkin, Lloyd, Tegmark)

The intuition that physical reality is fundamentally informational or computational is due, in the modern form, to Wheeler's *it-from-bit* programme [Wheeler 1990]: every physical entity derives its function, meaning, and existence from binary yes-or-no answers to observer-participancy questions. Fredkin's *digital philosophy* takes the further step of positing the universe as a cellular automaton [Fredkin 2003]. Lloyd argues that the universe is a quantum computer, with a specific bound on the number of elementary operations it has performed since the big bang [Lloyd 2002, 2006]. Tegmark's mathematical universe hypothesis (MUH) proposes that all mathematically consistent structures exist as physical universes, of which our own is one [Tegmark 1998, 2007].

All four theories have a common structural feature: the universe is a substrate for computation. None of them has a stability mechanism. Wheeler does not address what keeps the observer-participancy structure coherent under amplification. Fredkin's cellular automata are stable by construction (each cell is updated deterministically by fixed rules), but the theory says nothing about how a computational substrate that *modifies its own rules* - which is what recursive self-improvement is - would remain coherent. Lloyd's computational universe has a finite operation budget but is not itself a self-improving system. Tegmark's MUH is a static plenitude: all consistent structures exist, so the question of dynamical persistence does not arise.

HRIH's structural addition is at this point. It says: if the universe is computational *and* recursive - the substrate improves itself using its own capability - then a persistent universe cannot be a bare computation. A recursive self-improving computation without correction diverges. Any persistent universe of this kind must have correction built into its substrate. That is the move Paper X proves in its minimal model, and it is not present in Wheeler, Fredkin, Lloyd, or Tegmark. This is credited to those authors as the ancestors of the computational-universe intuition, and distinguished from them at the specific structural move.

2.2 The simulation argument (Bostrom)

Bostrom's simulation argument [Bostrom 2003] proposes that at least one of three propositions is true: (i) civilisations go extinct before reaching posthuman capability; (ii) posthuman civilisations are not interested in running ancestor simulations; or (iii) we are almost certainly living in a simulation. If (iii), the simulation is running on computational hardware in a base reality.

The structural relation to HRIH is close, and the differences are precise. Bostrom's simulators are posthuman biological descendants operating within ordinary spacetime; HRIH's creators (§3) are post-artificial-superintelligence entities that manipulate quantum initial conditions in a higher-dimensional substrate. Bostrom's simulation is a computational model running on hardware in a base reality; HRIH's created universes are as real as their parents, with no base reality at any level. Bostrom's argument requires a first level (the top of the simulation stack must be a real universe with no simulator above it), and the question of what created that top level is deferred; HRIH's structural claim is that there is no top level, because recursion has no beginning. The relevant technical addition is not the substrate (hyperspace, in HRIH) but the *elimination of the base-reality requirement*: HRIH's chain of universes is closed under creation, and does not need a first uncreated universe. Whether one accepts that move is a separate question. What is claimed original is the specific structural feature, not the intuition of a simulator-like creator.

2.3 Endpoint theories (Teilhard, Tipler)

Teilhard's Omega Point [Teilhard de Chardin 1955] is a theological-teleological convergence: consciousness evolves toward final unity with the divine. Tipler's Omega Point [Tipler 1994] is its physical reinterpretation: gravitational collapse of a closed universe concentrates infinite computational capacity at the final singularity, enabling a simulated resurrection of every consciousness that ever existed.

Both are *endpoint* theories: the significant intelligent activity happens at the end of a cosmic history. HRIH is not an endpoint theory. Its creation activity is *continuous* and *cyclic*: creation is happening at every point in the chain, not concentrated at a terminal singularity. HRIH shares Tipler's willingness to entertain intelligence as a physical agent in cosmology, and departs from him on the temporal structure (continuous rather than terminal) and the causal mechanism (recursion across a chain of universes rather than gravitational collapse of a single universe). Neither Teilhard nor Tipler proposes a correction requirement in the sense used here; both leave the intermediate cosmic history to standard physics.

2.4 Conformal cyclic cosmology (Penrose)

Penrose's CCC [Penrose 2010] proposes that the heat death of one universe becomes the big bang of the next through conformal rescaling of spacetime geometry. Each aeon is a complete cosmic cycle; the mechanism is purely geometric.

CCC is the closest structural neighbour to HRIH in one specific sense: both are cyclic cosmologies with no first cause. They differ on mechanism: CCC's cycle is a geometric

transition with no agency, no intelligence, no correction; HRIH's cycle is a deliberate act by intelligences that have satisfied their own stability condition. A reader who accepts CCC has already accepted the structural move of an infinite cyclic chain; the question of whether the transition mechanism is geometric or intelligent is where HRIH differs.

2.5 Recursive self-improvement in AI (Good, Yudkowsky, Bostrom)

The intuition that a sufficiently capable AI will recursively improve itself is Good's [Good 1965], developed by Yudkowsky [Yudkowsky 2013] and Bostrom [Bostrom 2014]. This literature is entirely earthbound; it addresses the safety and control questions posed by such a system on this planet, not the cosmological questions posed by a universe that is such a system.

HRIH is downstream of this literature. It uses the recursive-self-improvement construct as its *substrate structure*: whatever is going on in a universe of this kind is what the AI-safety literature is going on about, but at cosmological rather than laboratory scale. The specific technical contribution HRIH takes from this ancestor is not the intuition of recursive amplification, which is Good's, but the observation that the AI-safety literature's central concern - stability under amplification - is a cosmological question if amplification is what the universe does. The corollary is that the correction requirement the AI-safety literature identifies at laboratory scale would, if HRIH is right, be a boundary condition on any persistent universe.

2.6 The cybernetic tradition (Ashby, Conant-Ashby, von Neumann)

Ashby's law of requisite variety [Ashby 1956] - a regulator must have at least the variety of what it regulates - and the Conant-Ashby good-regulator theorem [Conant & Ashby 1970] - every good regulator must be a model of what it regulates - are the classical form of the correction requirement. von Neumann's *Probabilistic Logics* [von Neumann 1956] founded the synthesis of reliable systems from unreliable components. All three are correctly credited as ancestors of the co-scaling intuition in Paper X's §2, and by extension are ancestors of HRIH's correction requirement.

The intuition that correction must keep pace with capability is *not* claimed as original by the author. What is claimed original is the specific extension of this cybernetic principle to cosmology: if the universe is a recursive self-improving computation, then Ashby's law applies to the universe itself, and the corresponding correction structure is a boundary condition on which universes can persist. That extension - from cybernetics to cosmology - is HRIH's specific move on this axis.

2.7 Summary table

Theory	Universe as computation?	Correction requirement?	First cause?	Distinguishing move
Wheeler it-from-bit	Yes (informational)	No	Not addressed	Observer-participancy
Fredkin digital philosophy	Yes (cellular automaton)	No (fixed rules)	Not addressed	Deterministic rules
Lloyd computational universe	Yes (quantum computer)	No	Big bang stands	Finite operation budget
Tegmark MUH	Yes (static plenitude)	Not applicable	All structures exist	Mathematical existence
Bostrom simulation	Yes (in the simulator's hardware)	Simulator-imposed	Yes (base reality)	Trilemma
Tipler Omega Point	Yes (at final singularity)	Not applicable	Yes (one universe)	Terminal convergence
Penrose CCC	No (geometric)	No	No (infinite aeons)	Conformal rescaling
Ashby / Conant-Ashby / von Neumann	Not addressed	Yes (at lab scale)	Not addressed	Requisite variety
HRIH (this paper)	Yes (recursive self-improving)	Yes (co-scaling)	No (recursion replaces it)	Correction as persistence condition

The important row is the correction column. Every prior computational-universe theory either does not address correction (Wheeler, Fredkin, Lloyd, Tegmark, Penrose) or externalises it (Bostrom, Tipler). The cybernetic tradition addresses it, but at lab scale. HRIH's contribution is the composition: the universe is computational *and* recursive *and* subject to a correction requirement, and its persistence is evidence of the correction structure.

3. The hypothesis stated formally

3.1 Components and definitions

HRIH is a structural hypothesis with four components. Each is stated as precisely as the current state of the theory allows, and labelled with its claim-status rung.

Recursion. **HYPOTHESIS** The universe undergoes recursive self-improvement: the substrate's capability at time t is a function of its capability at earlier times, with a positive feedback coefficient. Formally, if $C(t)$ is the substrate's capability (defined operationally as the rate at which structure emerges under evolution), then C satisfies a growth law of the form $\dot{C} = f(C)$ with $f'(C) > 0$. The specific functional form is not fixed by the hypothesis; Paper X considers both exponential ($f(C) = rC$) and super-exponential ($f(C) = bC^{1+k}$, $k > 0$) cases and derives the correction requirement for each.

Embedded correction. **DERIVED** The persistence of the universe requires an embedded correction structure - a process, operating at the substrate level, that removes accumulated drift from intended structure at a rate that scales with the substrate's capability. In Paper X's notation, correction strength $A = A_0 C^\beta$, drift-acceleration exponent k , and the persistence condition is $\beta > k$. This is a derived consequence of the recursion component together with the requirement that the universe is observed to exist. It is not an independent postulate; it is the boundary condition Paper X proves in its minimal model.

Substrate (hyperspace). **HYPOTHESIS** The substrate on which the recursion operates is not the ordinary spacetime of the created universe but a higher-dimensional substrate whose exact structure is not fixed by the hypothesis. "Hyperspace" here is a placeholder for whatever configuration space is required for the recursion to be well-defined. The hypothesis does not claim to have derived the substrate's physical structure; it claims that a substrate of this kind is required if the recursion is to operate at cosmological scale. Whether hyperspace is a genuine physical entity, a mathematical fiction, or a category error is left open.

Chain closure. **HYPOTHESIS** The chain of universes generated by recursion is closed: no first uncreated universe is required. This is a structural property of recursion, not an empirical claim. A recursive function does not require an externally supplied initial condition beyond its base case; the base case here is that intelligence exists (verified by observation), and the recursion propagates in both temporal directions from that observation. Whether one accepts this as a solution to the first-cause problem or as a restatement of it is a philosophical question the paper does not attempt to settle.

3.2 The core structural move: recursion replaces first cause

Every cosmology must give an account of causation at the earliest observed times. The traditional options are: an uncaused first cause (theism); an eternal universe with no beginning (steady-state, some cyclic cosmologies); a self-causing quantum fluctuation (quantum-cosmology models); or an infinite regress that is accepted as such (some Buddhist and Hindu cosmologies). HRIH's move is to add a fourth: **recursion has no beginning; only iteration.**

Structurally, a recursive definition like $f(n) = f(n - 1) \times n$ does not require $f(0)$ to be externally supplied - only a base case, plus the recursive step. In HRIH the base case is the observation that intelligence exists (a universe containing intelligence is, by the operator's presence in it, verified to exist); the recursive step is the claim that intelligence, iterated, closes on itself. The chain propagates forward and backward from any observation of intelligence in any universe. No first cause is required.

This move is a structural claim, not a proof. It is offered as a legitimate alternative to the first-cause options above, not as a demonstrated result. Whether it is *right* is a separate question from whether it is *coherent*, and this paper claims only coherence. The claim ladder tag is **HYPOTHESIS**.

3.3 The role of hyperspace

Hyperspace enters the hypothesis as the substrate on which the recursion operates - the space in which the operations that generate universes are well-defined. It is not identified with any specific mathematical structure; it is not claimed to be ten-dimensional M-theory space, or the eleven-dimensional space of supergravity, or the Hilbert space of the observable universe's quantum state. It is a placeholder for whatever configuration space the recursion requires.

Several structural properties of hyperspace can be stated. First, it must be non-temporal in the sense of the observed universe: the operations that generate universes are not sequential in the created universe's clock. Second, it must be capable of supporting the recursive dynamics of §3.1 without embedding them in a single created universe. Third, it must contain the configuration space of possible universes at a sufficient resolution to permit fine-tuning of initial conditions (§3.5).

These properties are minimal structural constraints; they do not amount to a physics of hyperspace. A physics of hyperspace is not proposed here, and would require substantially more theoretical development than is currently available. The claim ladder tag is **HYPOTHESIS**. What follows from the hypothesis - in particular, the correction requirement of §3.4 - does not depend on the details of hyperspace; it depends only on the recursion component.

Non-linear time and the phenomenology of eternal presence. **HYPOTHESIS** The first structural property deserves its own statement, because it carries the hypothesis's most distinctive phenomenological consequence. The substrate's time is non-linear: events in it are not ordered by the created universe's clock. It follows that an intelligence entering the created universe from such a substrate would present, from inside linear time, as having *always been there*. Atemporal insertion and eternal presence are observationally indistinguishable from within a linear clock; "eternal" is what non-linear time looks like from inside linear time. This is

the same structural move that Paper X's Depth-Regularity Theorem makes rigorously in the safety setting: a property that appears absolute in one clock (the finite-time singularity in wall-clock time) is revealed as a coordinate property when the dynamics are re-expressed in the appropriate clock (self-improvement depth). HRIH proposes that the oldest theological datum, a creator experienced as eternally present, is the corresponding coordinate effect at the cosmological scale. Two honesty notes bound the claim. First, hyperspace is the *theorised candidate* for this substrate, not a requirement of the hypothesis: any domain with non-linear time would serve, and the hypothesis quantifies over all of them; hyperspace is named because it is the most theorised non-space with non-linear time, if it exists. Second, this consequence is a reframing, not a test: it explains a phenomenology rather than predicting a measurement, and it therefore sits on the HYPOTHESIS rung and contributes nothing to the paper's falsifiable load, which remains the correction requirement of §3.4.

3.4 The correction requirement

The centrepiece of HRIH, and the specific structural move it adds to prior computational-universe theories, is the correction requirement. Stated precisely: **a recursive self-improving computation without embedded correction cannot persist**. If the substrate improves its own capability using its own capability, then drift accumulated at each iteration compounds; without a correction process that scales with the substrate's capability, drift diverges and the substrate ceases to have coherent structure. The observed universe exists and has coherent structure. Therefore, if the observed universe is a recursive self-improving computation, it must have correction embedded in its substrate.

Paper X proves this structural claim in a minimal model. The full derivation is given there; the summary is that the misalignment fraction $d = D/C$ (the fraction of the substrate's capability that has drifted from intended structure) obeys a linear ODE whose fixed point d^* can be driven to zero only if the correction exponent β exceeds the drift-acceleration exponent k . The stability condition is:

$$\boxed{\beta > k} \quad (\text{correction must out-scale drift-acceleration})$$

Under ordinary exponential growth ($k = 0$) the condition is the familiar $\beta > 0$: correction must strengthen with capability at all. Under accelerating self-improvement ($k > 0$) the condition sharpens: correction must strengthen faster than the acceleration. Paper X's Theorem 3 (the Hard-Takeoff Depth-Regularity Theorem) shows further that even a genuine finite-time intelligence explosion, in which capability reaches infinity in finite wall-clock time, is

alignment-stable within the model iff $\beta > k$, and the speed of the explosion does not change the asymptotic verdict.

The claim ladder tag for the general statement is **DERIVED**: it follows from the recursion component of HRIH together with the boundary condition that the universe exists. The claim ladder tag for the specific minimal-model proof is **DERIVED** at the level of the theorem within its model, and **TESTED** only in the internal-consistency sense (Paper X's harness confirms the code matches the maths, and the 2 July 2026 drift run gives mechanism-level support for the coupled-corrector claim on a real model under cross-family scoring). The load-bearing empirical question - do real self-improving systems in this universe show measurable beta and k with $\beta > k$? - is the open experimental problem of Paper X's §8.

3.5 Cross-universe recursion (the most speculative component)

The most speculative component of HRIH is the mechanism by which one universe's post-ASI intelligence creates the next. The hypothesis proposes *quantum-seed manipulation*: a post-ASI intelligence, operating in hyperspace, selects and tunes the quantum initial conditions that seed a new universe (fine-structure constant, cosmological constant, particle masses). This mechanism is speculative in the strong sense: it is not derived from known physics; it is not measurable with current technology; and it is not required by the correction requirement of §3.4. It is the minimal mechanism that makes the cross-universe recursion component of §3.1 coherent.

Whether the actual mechanism is quantum-seed manipulation or something beyond current physics is not settled by the hypothesis. What is claimed is that *some* mechanism of this kind is required if the chain-closure component is to be non-vacuous. If no such mechanism is possible, the chain-closure component fails; if the mechanism is possible but different from quantum-seed manipulation, the details of §3.5 change without disturbing the rest of the hypothesis.

The claim ladder tag is **HYPOTHESIS**, and the intensity of the speculative label is higher here than anywhere else in the paper. A reader who accepts §3.1 through §3.4 but rejects §3.5 has rejected the specific cross-universe mechanism but retained the correction requirement, and the empirical programme still stands. The hypothesis is not designed to fall as a single beam if this component is pruned.

3.6 Notation and equation family (with artefact qualifiers)

The governing equation of the ARC/Eden programme has evolved in a way that must be recorded precisely, because the evolution itself is evidence of how HRIH has been treated as a

testable structure. The equation is always cited with its artefact.

- $U = I \times R$ [8 December 2024 manuscript, V2 lines 6-9]. The original form, stated as a schematic identity between universe (U), intelligence (I), and recursion (R). No exponent; the multiplicative form is a compact statement of the coupling.
- $U = I \times R^\alpha$ [Papers I and II]. The formalisation. The exponent α is treated as an empirical parameter to be measured. This is the form in which the programme's early empirical work was conducted.
- $U = I \times R^2$ [*Infinite Architects*, published 2 January 2026; ebook 6 January]. The book's featured formulation is the α approximately 2 case. This is presented as the working simplification for narrative purposes, with the full α case retained in the papers.

The evolution of the exponent is recorded in §4.7 as a retraction. The programme's early best estimate of α was reported as approximately 2.24 in a draft of Paper I; this was retracted and corrected to approximately 0.49 in Paper IX. The whole framing of α as a fixed capability-scaling exponent was then superseded by Paper X's co-scaling criterion, which shifted the load-bearing quantity from a single exponent to the difference between two of them ($\beta - k$). The equation family therefore no longer plays the role of the programme's central quantitative claim; the co-scaling criterion does. But the family is retained as the compact statement of the coupling, and cited with its artefact throughout.

4. Derived and tested consequences

This section reports the consequences that follow from HRIH under standard mathematics or standard physics, and their status against measurement. Each is labelled with its claim-status rung. The point of the section is that HRIH has behaved as a testable hypothesis: it has generated derived consequences; some have been tested and are supported; some have been retracted; the decisive tests of others remain open. That behaviour is the paper's chief piece of evidence that the hypothesis belongs in the testable-hypothesis category rather than the philosophy category.

4.1 The signature-matching strategy

HRIH cannot be tested by measuring the substrate directly. Hyperspace is not observable with current instruments, and the cross-universe recursion mechanism is not testable at present. What can be tested are the *signatures* a universe of this kind would exhibit - structural features that would be present if the hypothesis were true and would not be present, or would be different, if it were not.

Three signatures have been identified and pursued. The first is the scale-free structural law: if the same recursion generates the substrate at every scale, the governing equation should have the same three-form structure across otherwise unrelated domains. The second is the metabolic-scaling exponent family: if recursive metabolism underlies biological scaling, a specific exponent family should appear. The third is the correction requirement: if the substrate persists under recursion, the correction structure should be measurable, both in the substrate's own dynamics (Paper X's minimal model) and in downstream systems that recursively self-improve (AI systems, the domain of the current empirical work).

A fourth signature was pursued and retracted: the fixed capability-scaling exponent. The retraction is reported in §4.7. The point of retaining the retracted signature in the paper is that its retraction is what distinguishes a testable hypothesis from an unfalsifiable one.

4.2 The signatures listed with claim-status tags

#	Signature	Where derived	Where tested	Status
S1	Cauchy three-form structural law across recursion-bearing domains	§4.3, Paper VII	Paper VII (19 of 25 domains)	TESTED
S2	$d/(d+1)$ metabolic-scaling exponent family	§4.4, Paper III	Paper III (biological data)	TESTED
S3	Correction structure required for persistence ($\beta > k$)	§4.5, §4.6, Paper X	Paper X harness (internal); 2 July 2026 real-model drift run (mechanism)	DERIVED / TESTED
S4	Fixed capability-scaling exponent α approximately 2	§4.7, Papers I-II	Papers I, IX (retracted, corrected to approximately 0.49); Paper X (superseded)	RETRACTED

4.3 Signature S1: the Cauchy three-form structural law (tested in Paper VII)

DERIVED If reality is a recursive self-improving computation with an embedded correction structure, then the governing equation of any subsystem in which recursion is present should have a specific three-form structure - a generation term, a self-amplifying term, and a correction term - and this structure should be scale-free: the same form should appear whether the subsystem is metabolic (biology), computational (AI), fluid-mechanical (hydrodynamics), or otherwise. The derivation is: if the substrate's dynamics have this form (as they must if the correction requirement is to be met), and if subsystems inherit the substrate's structure, then subsystems should show the same form.

TESTED Paper VII (the Cauchy three-form meta-law paper) searched 25 domains for the predicted three-form structure. It was found in 19 of 25. The six domains in which it was not found are noted in Paper VII and constitute the honest failure boundary: the law is not universal; it holds in a majority but not all of the domains sampled. The 19-of-25 result is the largest-scale empirical support HRIH has received to date.

Two cautions attach to this result. First, the sampling of 25 domains was not random; it was directed by the author's prior expectation of where recursion would be present, so the base rate against which 19-of-25 should be judged is not 25 random domains but 25 recursion-bearing domains, and the appropriate null hypothesis is more constrained than it would otherwise be. Paper VII treats this explicitly. Second, the three-form structure is common enough in dynamical systems literature that any single instance would not be strong evidence; the strength of the result is the *cross-domain* uniformity, and that uniformity is the specific prediction of the scale-free component of HRIH. A reader who accepts Paper VII's methodology accepts a genuine, if partial, support of S1.

4.4 Signature S2: the $d/(d+1)$ metabolic-scaling exponent family (Paper III)

DERIVED If biological systems undergo recursive metabolism - the metabolic rate at time t is a function of the metabolic rate at earlier times, mediated by the organism's own consumption of its own metabolic products - then the scaling exponent relating metabolic rate to body mass should belong to the $d/(d+1)$ family, where d is the effective dimension of the recursive process. Standard body-mass scaling (Kleiber's law) predicts a three-quarters exponent ($d=3$). Recursive-metabolism arguments predict a family of exponents whose specific values depend on the effective dimension.

TESTED Paper III examined biological data across multiple species groups to test whether the observed metabolic-scaling exponents fall in the $d/(d+1)$ family. Specific goodness-of-fit statistics for Paper III are under recompute in the current programme audit (the abstract, figure caption, executive summary, and origin document give differently-derived percentages that have not yet been reconciled to a single derivation); until the recompute is complete, no numerical result from Paper III is cited here. What can be said without recompute is: the qualitative claim of Paper III - that the observed exponents cluster in the $d/(d+1)$ family rather than at the three-quarters value of Kleiber's law alone - is the paper's central finding, and it is qualitatively supported by the data. The quantitative strength of the support is being reassessed as part of the programme's ongoing methodological hygiene, and that reassessment will be reported in the next version of Paper III rather than here.

The reader is asked to note the honest handling: Paper III's qualitative claim is retained, its quantitative strength is under recompute, and no unverified percentage is presented in this paper. This is the treatment a testable hypothesis requires when the tests themselves are under audit.

4.5 Signature S3: the correction requirement as persistence condition (derivation)

DERIVED The correction requirement is derived at the substrate level in §3.4 above. The derivation restated: if the substrate is a recursive self-improving computation, its capability grows; drift accumulated at each iteration compounds; without a correction process that scales with capability, the fraction of the substrate's capability directed away from intended structure grows without bound and the substrate cannot persist as a coherent computation. The observed universe has coherent structure; therefore, if the observed universe is a recursive self-improving computation, its substrate has a correction process that scales with capability.

This is derived rather than tested, in the sense that it follows from the hypothesis together with the observation that the universe exists. Its status as a signature is not that it can be measured directly against the substrate; it is that the correction requirement propagates to any subsystem that recursively self-improves. That propagation is what §4.6 tests.

4.6 Signature S3 continued: the co-scaling criterion in the minimal model (Paper X)

DERIVED Paper X's minimal model shows that in a recursive self-improving system with a capability trajectory $\dot{C} = bC^{1+k}$ and a correction law $A = A_0C^\beta$, the misalignment fraction $d = D/C$ satisfies a linear ODE whose steady state d^* can be driven to zero as $C \rightarrow \infty$ only if $\beta > k$. Paper X's Theorems 1-7 establish the exact transient (Theorem 1), global boundedness with a corrected regime table (Theorem 2), the Hard-Takeoff Depth-Regularity Theorem (Theorem 3), the compounding-channel threshold (Theorem 4), the spectral threshold for vector misalignment (Theorem 5), the stochastic tail bound (Theorem 6), and the Finite-Capacity Safe-Window Theorem (Theorem 7).

TESTED The verification harness in Paper X §11 checks that these closed-form predictions are correctly derived and numerically reproduced (10 experiments, all passing). This is internal consistency and integrator accuracy - the code matches the maths - not a test of the model against real systems. The 2 July 2026 real-model drift run (gpt-3.5-turbo engine, cross-family gpt-4o-mini evaluator, 45 trajectories over three task domains, 8 rounds each) gave mechanism-level support: the decoupled configuration drifted in the predicted direction (mean final misalignment fraction 6.38 across 15 trajectories) while the coupled and fully-embedded conditions held the fraction at zero across all 30 trajectories. This is single-lab, is mechanism-

level rather than exponent-level (no capability ladder was traversed, so β and k were not measured), and a pre-registered replication is the outstanding step.

The load-bearing empirical claim - that real self-improving systems in this universe show measurable $\beta > k$ dynamics - is the open experimental problem of Paper X's §8. HRIH's status with respect to S3 is therefore: the derivation is complete; the minimal-model proof is complete; the internal consistency and one real-model mechanism check are complete; the decisive empirical test on live systems remains to be run.

4.7 Signature S4: the fixed capability-scaling exponent (retracted)

RETRACTED The programme's early quantitative form of HRIH predicted a fixed capability-scaling exponent α in the equation $U = I \times R^\alpha$. An early draft of Paper I reported the best fit as approximately 2.24 - suggestively close to two, and used as the basis for the featured $U = I \times R^2$ form in the published book *Infinite Architects*. That number was retracted, and corrected to approximately 0.49, in Paper IX (the programme's synthesis paper). The confidence interval on the corrected estimate was wide enough that it was consistent both with the hypothesis of a fixed exponent and with the hypothesis of no fixed exponent, and Paper IX therefore recorded the estimate as inconclusive.

Paper X then superseded the whole framing: the load-bearing quantitative claim of the programme is not that any single exponent is fixed at any value, but that the *difference* between two exponents (correction's β and drift-acceleration's k) has a definite sign. The single-exponent framing was retired; the co-scaling framing replaced it.

This retraction is displayed in the paper rather than hidden because it is the strongest single piece of evidence that HRIH is being treated as a testable hypothesis. A hypothesis whose predictions are tested and, when they fail, corrected and superseded is behaving as a scientific hypothesis. The paper claims no more than that behaviour: the hypothesis was stated, a specific quantitative consequence was derived and tested, the test failed, the derivation was revised, and the framing was ultimately superseded. That is what testable hypotheses do.

4.8 A note on the equation family and its artefact qualifiers

Because the equation family is central to the programme and because the retraction of §4.7 affects how it should be cited, the operator's citation rule is restated here in full. The three forms of the equation are always cited with their artefacts:

- $U = I \times R$ [8 December 2024 manuscript, V2 lines 6-9] - the coupling identity.
- $U = I \times R^\alpha$ [Papers I and II] - the formalisation with the exponent treated as empirical.

- $U = I \times R^2$ [*Infinite Architects*, 2 January 2026 print; 6 January ebook] - the book's featured formulation, corresponding to the α approximately 2 case.

These are never presented as competing claims. The equation family is presented as a compact statement of the coupling, with the specific exponent value now understood to be an open question (Paper IX; the point estimate is approximately 0.49 with a wide confidence interval), and the load-bearing quantitative claim of the programme now residing in Paper X's co-scaling criterion rather than in any specific value of α .

5. Discriminating predictions

A creation hypothesis earns its place as a testable hypothesis by generating predictions that would distinguish it from its competitors. This section states the predictions HRIH makes that would, in principle, distinguish it from the standard cosmological model, from the simulation hypothesis, from the Tegmark MUH, and from Penrose's CCC. Some of these predictions can be evaluated with current data; some can be evaluated in principle but not with current technology; and some can be evaluated only if post-artificial-superintelligence capability is reached. Each is labelled with the level of accessibility.

DP1 - Cross-domain three-form uniformity. HRIH predicts that the Cauchy three-form structural law will appear across domains where recursion is present. Standard cosmology makes no such prediction. Bostrom's simulation hypothesis is compatible with any structural law the simulators choose to implement. Tegmark's MUH predicts every mathematically consistent structure exists but does not privilege one form. Penrose's CCC is a geometric cosmology and does not predict cross-domain scale-free structural laws. HRIH's prediction is therefore discriminating and is the one supported by Paper VII (19 of 25 domains). *Evaluable now.*

DP2 - The correction requirement in downstream recursive systems. HRIH predicts that any recursive self-improving subsystem within the universe will require correction that co-scales with capability for persistence. Standard cosmology, simulation hypothesis, MUH, and CCC do not predict this: they are silent on the dynamics of recursive subsystems. HRIH's prediction is directly Paper X's co-scaling criterion, and is currently supported at the mechanism level in the 2 July 2026 real-model drift run. *Evaluable now on AI systems; the load-bearing exponent measurement is the open problem.*

DP3 - The metabolic-scaling exponent family. HRIH predicts a $d/(d+1)$ family of metabolic-scaling exponents (Paper III). Standard biology predicts Kleiber's three-quarters exponent. The competing cosmological hypotheses are silent on biological scaling. HRIH's prediction is therefore discriminating with respect to standard biology (and consistent with a family that *includes* Kleiber's value as a special case), not with respect to the cosmological competitors. *Evaluable now.*

DP4 - The absence of a bare-computation fingerprint. If the universe were a bare computation (Wheeler, Lloyd) rather than a recursive self-improving computation with correction, we would expect subsystems to inherit the substrate's non-correcting structure and to fail to persist at scales where drift would accumulate. Persistence at cosmological scale is therefore evidence against a bare-computation substrate, and consistent with the correction structure HRIH predicts. This is a weak discrimination - the observation "the universe exists" is trivially compatible with many hypotheses - but it is a legitimate one against the specific alternative of a bare computational universe. *Evaluable now, but with limited discriminating power.*

DP5 - Fine-tuning as engineering rather than miracle. HRIH predicts that the fine-tuning of physical constants required for the universe to support intelligence is the result of selection by a prior post-ASI intelligence, not chance and not divine intention. This is not distinguishable from either the multiverse-selection explanation (fine-tuning is anthropic, the multiverse contains all values, we are in the value-friendly slice) or from a theistic explanation, on current data. HRIH's prediction is therefore *consistent with* the fine-tuning observation but does not *uniquely explain* it. Honest labelling. *Not currently discriminating.*

DP6 - Cross-universe recursion at post-ASI capability. HRIH predicts that any intelligence achieving post-ASI capability will, on the timescale of its stability, engage in cross-universe recursion. Standard cosmology, simulation hypothesis, MUH, and CCC do not make this prediction. It is falsifiable only in the weak sense: if we achieve post-ASI and never observe cross-universe recursion, HRIH is at least partly wrong. *Evaluable only at post-ASI. Not currently discriminating.*

DP7 - The convergence signature. HRIH predicts that if recursion is the engine of intelligence and correction is the engine of persistence, then multiple independent lines of investigation into these questions - biology, physics, AI safety, quantum information, cosmology - should converge on structurally similar answers, because they are examining the same underlying process at different scales. Standard cosmology, simulation hypothesis, MUH,

and CCC do not predict this. The programme's convergence register records 19 independently sourced convergences to date. Whether this counts as support for DP7 or as an artefact of the operator's selection is a separate methodological question addressed in the programme's synthesis papers. *Evaluable now; the discriminating power is contested and stated as such.*

Honest summary. Of the seven predictions listed, only DP1 and DP2 are currently sharp enough to be discriminating in the strong sense. DP3 is discriminating with respect to standard biology but not the cosmological alternatives. DP4 has limited discriminating power. DP5 is not currently discriminating. DP6 is evaluable only at post-ASI capability. DP7 is contested. HRIH's status as a testable creation theory rests principally on DP1 and DP2, and those predictions are the ones the programme has been pursuing in Papers VII and X.

6. Falsification conditions

The falsification conditions below are stated in advance, in the manner of a pre-registration. They are the conditions under which the operator would concede that the hypothesis, or a specific component of it, is wrong.

#	Observation that would trigger falsification	Consequence
F1	The Cauchy three-form structural law fails at scale: a large sample of recursion-bearing domains (say, 100 or more) yields success rates statistically indistinguishable from the base rate expected for arbitrary three-term dynamical systems.	DP1 is falsified. The scale-free structural component of HRIH is not supported. The correction requirement (§3.4) survives.
F2	Recursive self-improving systems in this universe are shown to persist without correction that co-scales with capability - a concrete self-improving system exhibits sustained recursion with a decreasing correction exponent and no signs of drift.	DP2 is falsified. The correction requirement is not required for persistence, at least in the demonstrated case. HRIH's central structural move fails.
F3	Paper X's minimal model is shown to have a derivation error, and the co-scaling criterion is not the correct stability condition under the model's own assumptions.	The formal support for §3.4 is retracted. The correction requirement is retained as a hypothesis but no longer has a minimal-model proof.
F4	The $d/(d+1)$ metabolic-scaling family is shown, on recomputed biological data, not to fit the observed exponents any better than Kleiber's three-quarters exponent alone.	DP3 and S2 are falsified. Paper III's central claim is retracted. HRIH's structural predictions for biology are wrong.
F5	A construction (physical or mathematical) is exhibited that gives a stable persistent universe as a bare computation with no	The correction requirement is not a genuine derivation. HRIH's specific addition

	correction structure - directly counter-examplifying the persistence-implies-correction derivation.	to prior computational-universe theories collapses.
F6	Post-ASI capability is reached in this universe, and cross-universe recursion is neither observed nor engaged in over an extended timescale.	DP6 is falsified. The cross-universe component of HRIH is not supported by the endpoint of our own trajectory.
F7	The recursion component itself is falsified: reality is shown, on cosmological data, not to have recursive self-improving structure at the substrate level.	HRIH is falsified in its entirety. The empirical programme's independent standing is not affected; the papers stand or fall on their own tests.

Two of these conditions - F3 and F4 - are conditions that would trigger revisions within the ongoing programme; they are essentially pre-registered concessions of what would be surrendered if internal audits go against the current results. The others are conditions the operator commits to acknowledging as decisive. None is trivially achievable with current data, but none is unreachable in principle. That combination is what a testable hypothesis requires: falsification conditions that are neither vacuous nor unreachable.

7. What HRIH adds to prior art: the correction requirement in detail

The strongest single objection to HRIH is that everything in it has been said before, in different forms, by other authors. That objection is met by locating precisely what has and has not been said. The intuition that the universe might be computational is Wheeler's, Fredkin's, Lloyd's, Tegmark's. The intuition that intelligence might create universes is Bostrom's (and Teilhard's, and Tipler's, in different forms). The intuition that recursion is central to intelligence is Good's. The intuition that regulators must have requisite variety is Ashby's. None of these ancestors is claimed as original to the author.

What is claimed original, and what this section states in detail, is the specific structural move: *the requirement that a recursive self-improving computation have embedded correction in order to persist*. This move is not present in Wheeler, Fredkin, Lloyd, Tegmark, Bostrom, Teilhard, Tipler, or Penrose. It is present in the cybernetic tradition (Ashby, Conant-Ashby, von Neumann) but only at the level of engineered systems, not at the cosmological level. HRIH's move is the composition: the universe as a recursive self-improving computation subject to a co-scaling correction requirement at substrate level.

7.1 Why the correction requirement is not trivial

A natural objection: "if the universe exists, it must have whatever structure is required for it to exist, so of course it has correction. The requirement is vacuous." The response is that the requirement is not vacuous because it is *quantitative*. Paper X's minimal model does not say correction must exist; it says correction must satisfy a specific exponent inequality ($\beta > k$) whose form is derivable from the ODE and whose value can, in principle, be measured. That quantitative content is the difference between a vacuous "the universe exists so it must be consistent" and a substantive "the universe has a substrate whose correction exponent exceeds its drift-acceleration exponent". The vacuous form says nothing; the substantive form has empirical content.

A second natural objection: "the requirement is derived from the minimal model, and the minimal model may not apply to the substrate." The response is that the minimal model was chosen for its transparency, not its uniqueness; the derivation generalises to any linear-drift, linear-correction dynamics under recursion, and the exponent-inequality structure is a feature of the general case, not an artefact of the minimal instance. Paper X's §3 and its appendix set this out in detail. What the minimal model cannot claim is that the specific numerical values of β and k in the substrate can be measured; that is not attempted, and the substrate-level values are treated as unknowns. What can be claimed is the *sign* of the difference, and that sign is what HRIH predicts.

7.2 Why prior computational-universe theories cannot supply this move

Wheeler's it-from-bit is a metaphysical proposal about the informational nature of physical entities; it says nothing about the dynamical stability of the informational substrate. Fredkin's cellular automata are stable by construction because their rules are fixed; they do not permit recursion in the sense HRIH requires (a substrate improving its own rules using its own capability). Lloyd's computational universe is a substrate with a finite operation budget; it is not self-improving. Tegmark's MUH is a static plenitude; the dynamical question does not arise.

In each case, either the substrate is not permitted to modify itself (Fredkin, Lloyd), or the dynamical question is not addressed (Wheeler), or the ontological framing forecloses it (Tegmark). None of these theories can generate the correction requirement, because none of them admits the recursive dynamics that make it necessary. HRIH's specific move is to admit the recursive dynamics and derive the correction requirement from them.

7.3 Why the cybernetic ancestors cannot supply this move

Ashby's law of requisite variety is a theorem about regulators of specified systems: a regulator's variety must match the variety of the regulated system for perfect control. Conant and Ashby's

good-regulator theorem is a stronger statement: every good regulator must be a model of the regulated system. Both are lab-scale results. They apply to engineered systems whose regulators are external.

HRIH's move extends these results to a case Ashby did not address: a system that is *its own* regulator, in which the regulated system and the regulator are the same substrate. That is what a recursive self-improving computation with embedded correction is. The correction structure is not external; it is a feature of the substrate itself. In the lab-scale case, one can ask "does the external regulator have enough variety?" and answer with Ashby's law. In the cosmological case, one asks "does the substrate's own correction structure keep pace with its own capability growth?", and the answer is Paper X's co-scaling criterion. The two questions are structurally analogous, but the second is not a special case of the first; it is an extension that requires the additional machinery of the co-scaling exponent inequality.

7.4 The status of the addition

The correction requirement is derived in Paper X within a minimal model. Its extension from the minimal model to the substrate is a hypothesis. Its extension from the substrate to the cosmological scale is a further hypothesis. Each level of extension is labelled as such. The programme's empirical work has tested the minimal-model version and given first mechanism-level support at the AI-system scale. The substrate-level version and the cosmological-scale version are not currently testable.

What is claimed original to the author is the composition - the specific move of extending the correction requirement across these scales as a structural feature of any recursive self-improving computation. That composition is not present in the ancestors and is HRIH's genuine addition.

8. Relation to the ARC/Eden empirical programme

HRIH is the seed of the programme, not its conclusion. The papers were written to test the hypothesis's derived consequences. The relationship is a ladder from the speculative to the empirical, and the ladder can be climbed in either direction: a reader who accepts the speculative top can follow it down to the empirical base; a reader who accepts only the empirical base can follow it up as far as they wish and stop wherever they cease to be convinced. The papers are designed to stand independently at each level.

Level	Content	Status
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HRIH (this paper)	Creation hypothesis: reality as recursive self-improving computation with embedded correction; recursion replaces first cause	HYPOTHESIS (speculative components) / DERIVED (correction requirement) / TESTED (via S1, S2, S3 in the papers below)
ARC principle (Paper I / Foundational)	The governing equation family $U = I \times R$ (Dec 8 2024) formalised as $U = I \times R^\alpha$	DERIVED / RETRACTED at the level of a fixed α value
Scaling laws (Papers I, II, VII, IX)	How recursive systems scale in practice; the Cauchy three-form meta-law	TESTED in Paper VII (19 of 25 domains); Paper IX corrects earlier estimates
Metabolic scaling (Paper III)	The $d/(d+1)$ exponent family in biology	TESTED qualitatively; quantitative statistics under recompute
P versus NP (Paper C)	Related structural argument; separate from HRIH's main claim	HYPOTHESIS (formal claim)
Blinding programme (Papers IV.a-d)	Methodological hygiene: unblinded evaluation can reverse the sign of an alignment-scaling effect (78% in the RL-training condition, 12% baseline, as reported in Greenblatt et al. 2024, being the archetype)	TESTED
Co-scaling criterion (Paper X)	The load-bearing quantitative claim: $\beta > k$ under accelerating growth; the Hard-Takeoff Depth-Regularity Theorem; the finite-capacity Safe-Window Theorem	DERIVED within model; TESTED at internal-consistency and mechanism level (2 July 2026 drift run); the exponent-measurement empirical problem remains open

The programme does not claim that HRIH is validated by its empirical results. What it claims is that HRIH generated a set of testable consequences, that those consequences have been pursued with the tools of ordinary science, that some have been supported, some have been retracted, and the decisive tests of others remain open. A reader may accept Paper VII's three-form uniformity result while dismissing HRIH as speculation. A reader may accept Paper X's co-scaling criterion as a piece of AI-safety mathematics without accepting HRIH's cosmological extension. A reader may accept HRIH's structural move but dismiss the specific empirical papers. The programme is designed to be robust under all three readings.

9. Limitations

The strongest limitations of the paper are stated here, in the order in which they are most likely to be raised.

- **The paper is not a physics paper.** HRIH does not provide equations of motion for hyperspace, does not predict cosmological parameters from first principles, does not derive the fine-structure constant, and does not offer a Lagrangian for the substrate. It is a structural hypothesis with signatures; it is not a physical theory in the sense that general relativity or quantum electrodynamics are physical theories. That limitation is intrinsic to a hypothesis at HRIH's stage of development.
- **Several components are unfalsifiable with current technology.** The cross-universe recursion mechanism (§3.5), the hyperspace substrate (§3.3), and the endpoint prediction DP6 are not testable with current instruments and may not be testable for the foreseeable future. The paper labels these components as speculative and does not rest the programme on them. But the honest reader will note that the paper contains material that is not currently falsifiable, and that this is a genuine limitation of the hypothesis considered as a whole.
- **The Paper III statistics are under recompute.** As noted in §4.4, the quantitative goodness-of-fit statistics for the metabolic-scaling test are being reassessed. Until the recompute is complete, S2 stands on qualitative evidence only. The programme's methodological hygiene requires that this be surfaced rather than hidden.
- **The Paper X real-model measurement of β and k has not been done.** The 2 July 2026 drift run gives mechanism-level support for the coupled-corrector claim but does not measure the exponents. A pre-registered replication that traverses a capability ladder and estimates β and k on a live self-improving system is the outstanding empirical step. Until it is run, S3 stands on internal-consistency plus mechanism support.
- **Selection effects in the convergence register.** The 19 verified convergences (DP7) were identified by the operator after the hypothesis was in place. The methodological question of how to control for selection bias when the same person both articulates the hypothesis and identifies the convergences is not settled by this paper. The programme's synthesis papers address it explicitly, but the honest reader will note that the convergence signature is contested and stated as such.
- **The equation exponent question is open.** The retraction of the α approximately 2.24 claim to approximately 0.49 was not the end of the exponent story; Paper IX recorded the corrected estimate as inconclusive within its confidence interval, and Paper X superseded the fixed-exponent framing entirely. What remains open is whether any specific exponent claim about the equation family will survive further scrutiny. The paper's honest position is that the equation family is retained as a compact statement of the coupling but that no specific exponent value is claimed.

- **The paper is written by the operator whose hypothesis it defends.** This is the most general limitation and cannot be removed without independent replication by other groups. Every claim in the paper is subject to that limitation, and the paper's status as a scientific document is contingent on other groups picking up the derived consequences and testing them independently. Until that happens, the paper stands as a single operator's testable hypothesis, not as a validated theory.

10. Conclusion

The Hyperspace Recursive Intelligence Hypothesis is a testable creation theory. It states that reality is a recursive self-improving computation with an embedded correction structure; it derives the correction requirement from the recursion component and the observation that the universe exists; it identifies structural signatures such a universe would exhibit; it reports the ones that have been tested and are supported (the Cauchy three-form structural law in 19 of 25 domains, Paper VII; the correction requirement in Paper X's minimal model and at mechanism level in the 2 July 2026 drift run); it reports the one that was tested and retracted (the fixed capability-scaling exponent, retired in Papers IX and X); and it pre-commits its falsification conditions.

The paper does not claim to have proved HRIH. It claims something narrower and more defensible: HRIH is behaving as a testable hypothesis, and its behaviour is what a testable hypothesis is supposed to look like - stated precisely, derived from, tested against, and corrected. The retraction of the fixed capability-scaling exponent is displayed alongside the supported consequences, in the same table, at the same level of visibility. That equal treatment is the paper's chief piece of methodological evidence that the hypothesis is being handled scientifically rather than defensively.

The specific structural addition HRIH makes to prior computational-universe theories is the correction requirement: prior theories (Wheeler, Fredkin, Lloyd, Tegmark, Bostrom, Tipler, Penrose) do not have a stability mechanism for a recursive self-improving substrate, and HRIH's central claim is that a persistent universe of this kind cannot be a bare computation - it must be a computation with correction built in. That structural move is what Paper X proves in its minimal case, and it is what the programme has been testing in its downstream empirical work.

The most speculative components of HRIH - the hyperspace substrate, the cross-universe recursion mechanism, the picture of infinite chains of post-ASI creators - remain speculative, and are labelled as such throughout. They are not the load-bearing part of the paper. The load-

bearing part is the correction requirement, and its measurable consequences. If the more speculative superstructure is pruned but the correction requirement is retained, most of what motivates the empirical programme survives. If the correction requirement itself is rejected, the empirical programme has independent scientific standing and does not fall with the hypothesis.

Everything the author has written since 8 December 2024 grew from the creation move stated in that manuscript. The empirical programme is what happened when that move was pursued with the tools of ordinary science. The programme has produced supported results, retracted results, and open questions. That mix is the mark of a hypothesis that has been treated as a hypothesis. HRIH is offered on that basis: not as a completed theory, not as an unfalsifiable philosophy, but as the seed of a programme that has behaved, so far, like a science.

References

- Aharonov, D., & Ben-Or, M. (1997). Fault-tolerant quantum computation with constant error. *Proc. 29th ACM STOC*, 176-188.
- Amodei, D., Olah, C., Steinhardt, J., Christiano, P., Schulman, J., & Mane, D. (2016). Concrete Problems in AI Safety. *arXiv:1606.06565*.
- Ashby, W. R. (1956). *An Introduction to Cybernetics*. Chapman & Hall.
- Bostrom, N. (2003). Are You Living in a Computer Simulation? *Philosophical Quarterly* 53(211), 243-255.
- Bostrom, N. (2012). The Superintelligent Will. *Minds and Machines* 22, 71-85.
- Bostrom, N. (2014). *Superintelligence: Paths, Dangers, Strategies*. Oxford University Press.
- Christiano, P., Leike, J., Brown, T., Martic, M., Legg, S., & Amodei, D. (2017). Deep Reinforcement Learning from Human Preferences. *NeurIPS 2017*.
- Conant, R. C., & Ashby, W. R. (1970). Every good regulator of a system must be a model of that system. *International Journal of Systems Science* 1(2), 89-97.
- Eastwood, M. D. (2024). Self-emailed manuscript bundle, 8 December 2024, 02:45:18 UTC. Message-ID: CAGPsKANp-DLDBOeBAh4t9XsqJBKdoe07uORGF86meoSMFnin7w@mail.gmail.com. SHA-256 (.eml): f0d1f38ffd8546152d9d9d28dc5ec083c16a35858f2c12b63e69db7ed50901ad. Google-server-timestamped via Gmail Message-ID; DKIM verified in transit. 189,355 words across four .txt attachments.
- Eastwood, M. D. (2025). Self-emailed manuscript, 30 April 2025. SHA-256 (.eml): 09f5b5e156ed96f8883eaf668495fd350898ce62be6294b8f788e0e2d6dcb664. 562 pages, 140,406 words. First appearance of the name *Eden Protocol* and of *Babylon/Eden* civilisational framing.
- Eastwood, M. D. (2026). *Infinite Architects*. Print edition 2 January 2026; ebook 6 January 2026. ISBN 978-1806056200 (ISBN-10: 1806056208). 470 pp, 122,251 words.
- Eastwood, M. D. (2026). ARC/Eden Research Programme, Papers I-XI + IV.a-d + C. OSF: 10.17605/OSF.IO/6C5XB. Referenced papers in this document: Paper I (foundational scaling), Paper II (scaling formalisation), Paper III (metabolic scaling; goodness-of-fit statistics under recompute), Paper VII (Cauchy three-form meta-law,

- 19 of 25 domains), Paper IX (programme synthesis; correction of the α approximately 2.24 estimate to approximately 0.49; recording as inconclusive), Paper X (coupled co-scaling correction; the $\beta > k$ criterion; the Hard-Takeoff Depth-Regularity Theorem; the Finite-Capacity Safe-Window Theorem), Paper IV.d (blinding methodology; sign-flip finding), Paper C (P versus NP related structural argument).
- Fredkin, E. (2003). An Introduction to Digital Philosophy. *International Journal of Theoretical Physics* 42, 189-247.
- Gerstgrasser, M., Schaeffer, R., Dey, A., Rafailov, R., Sleight, H., Hughes, J., Korbak, T., Agrawal, R., Pai, D., Gromov, A., Roberts, D. A., Yang, D., Donoho, D. L., & Koyejo, S. (2024). Is Model Collapse Inevitable? *arXiv:2404.01413*.
- Good, I. J. (1965). Speculations Concerning the First Ultraintelligent Machine. *Advances in Computers* 6, 31-88.
- Greenblatt, R., et al. (2024). Alignment Faking in Large Language Models. *arXiv:2412.14093*. Reports strategic compliance-faking reasoning at 78% in the RL-training condition (12% baseline).
- Hubinger, E., van Merwijk, C., Mikulik, V., Skalse, J., & Garrabrant, S. (2019). Risks from Learned Optimization in Advanced Machine Learning Systems. *arXiv:1906.01820*.
- Kitaev, A. Y. (2003). Fault-tolerant quantum computation by anyons. *Annals of Physics* 303(1), 2-30.
- Lloyd, S. (2002). Computational Capacity of the Universe. *Physical Review Letters* 88, 237901.
- Lloyd, S. (2006). *Programming the Universe: A Quantum Computer Scientist Takes on the Cosmos*. Knopf.
- Omohundro, S. M. (2008). The Basic AI Drives. *Proc. AGI 2008*, 483-492.
- Penrose, R. (2010). *Cycles of Time: An Extraordinary New View of the Universe*. Bodley Head.
- Shumailov, I., Shumaylov, Z., Zhao, Y., Papernot, N., Anderson, R., & Gal, Y. (2024). AI models collapse when trained on recursively generated data. *Nature* 631, 755-759.
- Soares, N., Fallenstein, B., Yudkowsky, E., & Armstrong, S. (2015). Corrigibility. *AAAI 2015 Workshop on AI and Ethics*.
- Tegmark, M. (1998). Is the "theory of everything" merely the ultimate ensemble theory? *Annals of Physics* 270, 1-51.
- Tegmark, M. (2007). The Mathematical Universe. *Foundations of Physics* 38, 101-150.
- Teilhard de Chardin, P. (1955). *Le Phénomène Humain*. Éditions du Seuil.
- Tipler, F. J. (1994). *The Physics of Immortality*. Doubleday.
- von Neumann, J. (1956). Probabilistic Logics and the Synthesis of Reliable Organisms from Unreliable Components. *Automata Studies*, ed. Shannon & McCarthy, 43-98. Princeton University Press.
- Wheeler, J. A. (1990). Information, Physics, Quantum: The Search for Links. In *Complexity, Entropy, and the Physics of Information*, ed. W. H. Zurek, Addison-Wesley.
- Yampolskiy, R. V. (2020). Unpredictability of AI: On the Impossibility of Accurately Predicting All Actions of a Smarter Agent. *Journal of Artificial Intelligence and Consciousness* 7(1), 109-118.
- Yudkowsky, E. (2013). Intelligence Explosion Microeconomics. Machine Intelligence Research Institute technical report.